ENTERPRISE PERSISTENCE ARCHITECTURE

DATABASE DESIGN & SCHEMA REFERENCE SPECIFICATION

Edu Track: Unified Data Architecture

High-Performance Document Modeling with Relational Integrity Constraints & Dual-Store Autonomous Synchronization

Database Engine:	MongoDB Atlas (M0/M10 Replica Set) + Local JSON Fallback Store
Data Modeling Approach:	Hybrid Document-Relational with Foreign Keys & In-Memory Indices
Collection Count:	16 Structured Production Collections
ACID Atomicity:	Document-level atomicity in MongoDB; Atomic fs.rename in JSON mode
Index Strategy:	Compound indices on {dept, sem, section}, {studentId, examId}, etc.

Architectural Overview: *Edu Track utilizes a document-oriented persistence paradigm modeled to ensure ultra-fast read latency for academic rosters, timetables, and real-time assessments while enforcing relational constraints. Every entity is modeled as a primary collection with cross-collection referencing via foreign keys (*`studentId, facultyId, subjectId, sectionId`*). A dual-store persistence manager mirrors every transaction across MongoDB Atlas and atomic disk-backed JSON stores, guaranteeing zero data loss during network disconnects.*

1. Entity Relationship & Cardinality Overview

The schema models the institutional hierarchy starting from Departments down to individual Student Exam Submissions:

Core Relational Cardinality:
- **Department (1) ---> (N) Subjects:** Each department owns multiple curriculum subjects across semesters.
- **Department (1) ---> (N) Sections:** A department contains multiple cohort sections (e.g. CSE-A, CSE-B).
- **Section (1) ---> (N) Students:** Students belong to exactly one department, semester, and section.
- **Faculty (1) ---> (N) Faculty Assignments:** Faculty are assigned specific subjects and sections.
- **Section (1) ---> (N) Class Timetables:** Cohort timetable entries mapping day, period, room, subject, and instructor.
- **Department (1) ---> (N) Exam Timetables:** Semester examination dates linking department subjects to halls.
- **Faculty + Section (1) ---> (N) Attendance Records:** Daily sessions recording student presence.
- **Faculty + Section (1) ---> (N) Course Materials:** Section-scoped documents and study notes.
- **Faculty + Section (1) ---> (N) Online Exams:** MCQs with total/pass marks, timer, and state machine.
- **Student + Online Exam (1) ---> (1) Exam Attempt:** Single-attempt record with score, tab switches, and answers.

2. Complete Collection Schema Reference

Collection: users — Central user identities (Admins, Faculty, Students) with role credentials

Field Name	BSON / JSON Type	Required	Default / Constraints	Description & Foreign Key
`_id / id`	ObjectId / String	**YES**	UUID v4	Primary key for authentication and cross-referencing
`email`	String	**YES**	Unique, Lowercase	Unique login email address across the institution
`passwordHash`	String	**YES**	bcrypt(10)	Securely hashed user password
`fullName`	String	**YES**	None	Legal full name of user
`role`	String	**YES**	'ADMIN' | 'FACULTY' | 'STUDENT'	RBAC authorization role
`department`	String	**OPTIONAL**	null for admin	Department code (e.g., 'Computer Science & Engineering')
`semester`	Number	**OPTIONAL**	null for faculty	Current academic semester (1 through 8)
`section`	String	**OPTIONAL**	null for admin	Enrolled or assigned section (e.g. 'CSE-A', 'CSE-B')
`rollNumber`	String	**OPTIONAL**	Unique for students	Official institutional roll / register number
`academicCgpa`	Number	**OPTIONAL**	0.00	Cumulative Grade Point Average (Students only)
`createdAt`	Date / ISOString	**YES**	Date.now()	Account provisioning timestamp

Collection: departments — Academic institutional departments and administrative faculties

Field Name	BSON / JSON Type	Required	Default / Constraints	Description & Foreign Key
`id`	String	**YES**	Auto-generated	Unique department identifier
`name`	String	**YES**	Unique	Full department title (e.g., Computer Science & Engineering)
`code`	String	**YES**	Unique, Uppercase	Short uppercase department code (e.g., 'CSE')
`hodName`	String	**YES**	None	Name of Head of Department
`totalSemesters`	Number	**YES**	8	Duration of academic curriculum in semesters
`activeSections`	Array	**YES**	['A', 'B']	List of recognized section letters
`createdAt`	Date / ISOString	**YES**	Date.now()	Creation timestamp

Collection: subjects — Department curriculum subjects, course codes, and credit hours

Field Name	BSON / JSON Type	Required	Default / Constraints	Description & Foreign Key
id	String	YES	Auto-generated	Primary subject key
code	String	YES	Unique per Dept	Official course code (e.g., 'CS401', 'CS402')
title	String	YES	None	Subject title (e.g., 'Database Internals & Storage Engines')
department	String	YES	FK -> departments.name	Owning department
semester	Number	YES	1-8	Academic semester level
credits	Number	YES	3	Academic credit units
type	String	YES	'THEORY' | 'LAB'	Instructional category

Collection: sections — Instructional student cohorts and section allocations

Field Name	BSON / JSON Type	Required	Default / Constraints	Description & Foreign Key
id	String	YES	Auto-generated	Unique section identifier
name	String	YES	None	Section code name (e.g. 'CSE-A', 'CSE-B')
department	String	YES	FK -> departments.name	Parent academic department
semester	Number	YES	1-8	Associated semester level
capacity	Number	YES	60	Maximum enrolled student capacity
facultyAdvisor	String	OPTIONAL	FK -> users.id	Faculty coordinator assigned to this section

Collection: faculty_assignments — Mapping instructors to specific departmental subjects and cohorts

Field Name	BSON / JSON Type	Required	Default / Constraints	Description & Foreign Key
id	String	YES	Auto-generated	Assignment record identifier
facultyId	String	YES	FK -> users.id	Reference to the teaching faculty account
facultyName	String	YES	None	Cached instructor name for fast lookup
subjectId	String	YES	FK -> subjects.id	Assigned subject reference
subjectCode	String	YES	FK -> subjects.code	Cached course code
department	String	YES	FK -> departments.name	Target department
semester	Number	YES	1-8	Target semester
section	String	YES	None	Designated teaching section (e.g. 'CSE-B')

Collection: class_timetables — Weekly instructional timetable matrix with room & period allocations

Field Name	BSON / JSON Type	Required	Default / Constraints	Description & Foreign Key
id	String	YES	Auto-generated	Unique period slot entry
dayOfWeek	String	YES	'Monday'..'Saturday'	Instructional day
periodNumber	Number	YES	1 to 8	Daily period index
startTime	String	YES	HH:MM format	Period commencement time
endTime	String	YES	HH:MM format	Period conclusion time
department	String	YES	FK -> departments.name	Target cohort department
semester	Number	YES	1-8	Target cohort semester
section	String	YES	None	Target section (e.g., 'CSE-A')
subjectCode	String	YES	FK -> subjects.code	Course subject code
subjectName	String	YES	None	Full course name
facultyId	String	YES	FK -> users.id	Assigned faculty member ID
facultyName	String	YES	None	Assigned faculty member name
room	String	YES	None	Physical or virtual lecture room / lab (e.g. 'Hall 101')

Collection: exam_timetables — University semester examination schedules with room venues and timings

Field Name	BSON / JSON Type	Required	Default / Constraints	Description & Foreign Key
id	String	YES	Auto-generated	Unique examination schedule entry
department	String	YES	FK -> departments.name	Department subject to the examination
semester	Number	YES	1-8	Semester level
subjectCode	String	YES	FK -> subjects.code	Examined subject code
courseTitle	String	YES	FK -> subjects.title	Official course title
examDate	String	YES	YYYY-MM-DD	Date of scheduled examination
startTime	String	YES	HH:MM format	Examination start time
endTime	String	YES	HH:MM format	Examination end time
room	String	YES	None	Examination venue / hall (e.g. 'Exam Hall 204')
instructions	String	OPTIONAL	University rules	Candidate guidelines and required stationery
createdBy	String	YES	FK -> users.id	Administrator ID who scheduled the exam

Collection: attendance_records — Daily session attendance tracking locked to calendar date

Field Name	BSON / JSON Type	Required	Default / Constraints	Description & Foreign Key
id	String	YES	Auto-generated	Unique attendance session identifier
sessionDate	String	YES	Current Date (YYYY-MM-DD)	Session date (locked strictly to current day)
department	String	YES	FK -> departments.name	Cohort department
semester	Number	YES	1-8	Cohort semester
section	String	YES	None	Target instructional section
subjectId	String	YES	FK -> subjects.id	Associated subject
facultyId	String	YES	FK -> users.id	Instructor who logged attendance
records	Array	YES	[]	Array of [{ studentId, rollNumber, status: 'present'|'absent' }]
submittedAt	Date / ISOString	YES	Date.now()	Server commit timestamp

Collection: course_materials — Section-scoped digital study materials, lecture slides and notes

Field Name	BSON / JSON Type	Required	Default / Constraints	Description & Foreign Key
id	String	YES	Auto-generated	Unique document identifier
title	String	YES	None	Document title / topic description
description	String	OPTIONAL	None	Brief notes or instructions from faculty
subjectId	String	YES	FK -> subjects.id	Related academic subject
department	String	YES	FK -> departments.name	Associated department
semester	Number	YES	1-8	Semester level
section	String	YES	None	Section privacy scope (e.g. 'CSE-B')
fileUrl	String	YES	None	Relative or cloud path to uploaded file
fileName	String	YES	None	Original uploaded filename
uploadedBy	String	YES	FK -> users.id	Faculty creator ID
uploadedAt	Date / ISOString	YES	Date.now()	Publication timestamp

Collection: assignments — Coursework assignments, project tasks, and lab assignments

Field Name	BSON / JSON Type	Required	Default / Constraints	Description & Foreign Key
id	String	YES	Auto-generated	Assignment identifier
title	String	YES	None	Assignment title and task summary
description	String	YES	None	Full coursework problem statement
department	String	YES	FK -> departments.name	Target department

`semester`	Number	**YES**	1-8	Target semester
`section`	String	**YES**	None	Target section
`subjectId`	String	**YES**	FK -> subjects.id	Curriculum course reference
`dueDate`	String	**YES**	YYYY-MM-DD	Final submission cutoff date
`totalMarks`	Number	**YES**	20	Maximum gradable marks

Collection: student_submissions — Student assignment uploads, submission text, and faculty grades

Field Name	BSON / JSON Type	Required	Default / Constraints	Description & Foreign Key
id	String	YES	Auto-generated	Unique submission record
assignmentId	String	YES	FK -> assignments.id	Associated coursework task
studentId	String	YES	FK -> users.id	Submitting student account
studentName	String	YES	None	Cached student full name
rollNumber	String	YES	None	Student roll number
submissionText	String	OPTIONAL	None	Student solution narrative or git repo link
fileUrl	String	OPTIONAL	None	Attached solution document path
submittedAt	Date / ISOString	YES	Date.now()	Actual timestamp of student upload
status	String	YES	'SUBMITTED' | 'GRADED'	Workflow progression state
marksAwarded	Number	OPTIONAL	null until graded	Evaluation score awarded by faculty
feedback	String	OPTIONAL	None	Evaluator feedback notes

Collection: online_exams — Time-bounded multiple-choice online examinations and quiz specifications

Field Name	BSON / JSON Type	Required	Default / Constraints	Description & Foreign Key
id	String	YES	Auto-generated	Unique online examination key
title	String	YES	None	Examination title (e.g. 'Lab Exam', 'Quiz 1')
instructions	String	YES	None	Candidate rules and proctoring guidelines
department	String	YES	FK -> departments.name	Cohort department
semester	Number	YES	1-8	Semester level
section	String	YES	None	Section scope ('All' or specific section)
subjectId	String	YES	FK -> subjects.id	Associated subject
subjectCode	String	YES	None	Cached course code (e.g. 'CS402')
examDate	String	YES	YYYY-MM-DD	Scheduled examination date
startTime	String	YES	HH:MM format	Window opening time
endTime	String	YES	HH:MM format	Window closing time
durationMinutes	Number	YES	60	Allowed test duration once student begins
totalMarks	Number	YES	20	Total aggregate marks
passMarks	Number	YES	12	Minimum threshold to pass
negativeMarking	String	YES	'None' | '0.25' | '0.5'	Penalty deduction per wrong answer
questions	Array	YES	[]	Array of [{ id, questionText, options: [4 strings], correctOptionIndex, marks }]
status	String	YES	'SCHEDULED'|'AVAILABLE'|'CLOSED'	Active lifecycle status
resultsReleased	Boolean	YES	false	Faculty flag granting student access to scores & answer key

Collection: exam_attempts — Recorded student test attempts, tab-switch infractions, and scores

Field Name	BSON / JSON Type	Required	Default / Constraints	Description & Foreign Key
id	String	YES	Auto-generated	Unique attempt identifier
examId	String	YES	FK -> online_exams.id	Target online examination
studentId	String	YES	FK -> users.id	Student who sat for the assessment
studentName	String	YES	None	Cached student name
rollNumber	String	YES	None	Cached student roll number
answers	Array	YES	[]	Recorded selections: [{ questionId, selectedOptionIndex }]
score	Number	YES	0	Evaluated marks scored
totalMarks	Number	YES	None	Maximum attainable marks

`percentage`	Number	**YES**	None	Score percentage (0.00% to 100.00%)
`isPassed`	Boolean	**YES**	false	True if score >= passMarks
`startedAt`	Date / ISOString	**YES**	Date.now()	Timestamp student clicked 'Attend Examination'
`submittedAt`	Date / ISOString	**YES**	Date.now()	Timestamp student completed or timed out
`tabSwitchCount`	Number	**YES**	0	Total proctoring window blur / tab-switch infractions
`status`	String	**YES**	'EVALUATED'	Completion state

Collection: marks_records — Official university semester marks, internal assessments, and grade points

Field Name	BSON / JSON Type	Required	Default / Constraints	Description & Foreign Key
id	String	YES	Auto-generated	Unique academic grade record
studentId	String	YES	FK -> users.id	Enrolled student account
subjectId	String	YES	FK -> subjects.id	Target subject
semester	Number	YES	1-8	Academic semester
internalMarks	Number	YES	Max 40	Continuous internal assessment score
externalMarks	Number	YES	Max 60	End-semester university theory/lab score
totalMarks	Number	YES	Max 100	Composite final mark
letterGrade	String	YES	'O','A+','A','B+','B','RA'	Institutional letter grade
gradePoints	Number	YES	10 to 0 scale	Numerical grade points for CGPA computation

Collection: notifications — System alerts, timetable changes, exam announcements, and grade alerts

Field Name	BSON / JSON Type	Required	Default / Constraints	Description & Foreign Key
id	String	YES	Auto-generated	Unique notification identifier
recipientId	String	OPTIONAL	FK -> users.id	Direct user recipient (null if cohort broadcast)
targetRole	String	OPTIONAL	'STUDENT'|'FACULTY'|'ALL'	Target role for wide announcements
department	String	OPTIONAL	None	Department filter for broadcast
section	String	OPTIONAL	None	Section filter for broadcast
title	String	YES	None	Notification title banner
message	String	YES	None	Full body text of alert
isRead	Boolean	YES	false	Read status tracking for user inbox
createdAt	Date / ISOString	YES	Date.now()	Dispatch timestamp

Collection: audit_logs — Security audit trail tracking administrative actions and system events

Field Name	BSON / JSON Type	Required	Default / Constraints	Description & Foreign Key
id	String	YES	Auto-generated	Audit event log identifier
userId	String	YES	FK -> users.id	Executing user ID
userRole	String	YES	Role claim	Role of user at time of action
action	String	YES	Uppercase Verb	Action type (e.g. 'CREATE_EXAM', 'RELEASE_RESULTS')
entity	String	YES	Collection name	Affected domain entity
entityId	String	OPTIONAL	None	Primary key of mutated entity
ipAddress	String	YES	Client IP	Originating client network address
details	String	OPTIONAL	None	JSON payload metadata describing the modification
timestamp	Date / ISOString	YES	Date.now()	Immutable event timestamp

3. Indexing Strategy & Performance Rules

To satisfy sub-50ms query latency, compound and unique indices are defined across high-frequency access paths:

Collection	Index Key Specification	Type / Constraint	Query Optimization Objective
users	{ email: 1 }	**Unique Index**	Ultra-fast credential lookup during authentication.
users	{ department: 1, section: 1 }	**Compound Index**	Fast retrieval of section student rosters.
class_timetables	{ department: 1, semester: 1, section: 1 }	**Compound Index**	Instant weekly timetable rendering.

`class_timetables`	`{ dayOfWeek: 1, periodNumber: 1, room: 1 }`	**Compound Index**	Conflict engine classroom collision checks.
`online_exams`	`{ department: 1, semester: 1, section: 1 }`	**Compound Index**	Listing assessments relevant to student cohort.
`exam_attempts`	`{ examId: 1, studentId: 1 }`	**Unique Compound**	Guarantees single attempt rule enforcement.
`attendance_records`	`{ sessionDate: 1, section: 1, subjectId: 1 }`	**Compound Index**	Retrieving daily section attendance logs.